

JAYPEE INSTITUTE OF INFORMATION TECHNOLOGY, NOIDA

B.TECH VI SEMESTER

REPORT FOR

Introduction to Block-chain Technologies



DECENTRALISED FILE SHARING PLATFORM

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Problem Statement

The process began with an analysis of the current state of file sharing. Recognizing the challenges associated with centralized servers, such as security and privacy issues, was a crucial step. This identification laid the foundation for exploring alternative solutions. The research delved into the security and privacy challenges posed by centralized file-sharing methods. Centralized systems are vulnerable to single points of failure, such as server downtime or Cyber attacks on a central repository. In decentralized systems, files are distributed across a network, reducing the impact of such failures.

MOTIVATION

The primary motivation for addressing the file-sharing problem was the pressing need for enhanced security and privacy in digital content distribution. The vulnerabilities of centralized systems underscored the importance of finding a decentralized solution.

The motivation extended to the desire for innovation in the field of file sharing. The research aimed to explore cutting-edge technologies, such as block-chain and IPFS, to revolutionize the existing methods and overcome the challenges faced by traditional file-sharing approaches. Seeking an efficient and reliable file-sharing platform was a driving force. The motivation stemmed from the aspiration to create a system that not only addressed security concerns but also provided a more efficient and reliable means of sharing digital content.

Functional Requirements

User Registration and Authentication :-

Users shall be able to register with the platform, providing necessary information. Authentication mechanisms, including secure password storage or alternative methods, shall be implemented to ensure user identity and prevent unauthorized access.

File Uploading and Sharing :-

Users shall be able to upload files to the platform securely. Files shall be stored on IPFS for decentralized distribution. Users shall have full control over file sharing permissions, including public, private, and restricted access options.

Decentralized Storage :-

The platform shall utilize a decentralized storage system to distribute files across the network. Incentive mechanisms, possibly based on block-chain tokens, shall be implemented to encourage users to contribute storage space.

Block-chain Integration :-

Smart contracts on a block-chain shall be employed for managing user permissions, file access, and incentives. The platform shall leverage block-chain for secure and transparent transaction history, ensuring accountability.

User Interface :-

An intuitive and user-friendly interface shall be developed to enable easy navigation and file management. Users shall be provided with real-time updates on file transfers, storage usage, and any platform related activities.